

Real Estate Price Prediction AI

(A Machine Learning Platform for Transparent Property Valuation in Jammu)

Abstract:- Property pricing in a market like Jammu is largely a matter of trust rather than evidence. A buyer is usually quoted by a broker with a stake in the deal, checks it against one or two nearby listings that may or may not be genuinely comparable, and makes a six or seven figure decision on that basis. There is no independent, data-backed reference point in the process at all. This project builds that reference point. Real Estate Price Prediction AI takes a property's details - locality, size, type, age, builder, infrastructure quality, legal status and returns a predicted price, an expected range, and an investment score, generated from an ensemble of four machine learning models trained on verified transaction data rather than opinion. The same platform lets a user look at how different Jammu localities are trending, weigh one property against several others directly, and see the reasoning the model used to arrive at its number, rather than accepting a figure at face value. Built in Python with Streamlit, the system runs on 1,200 verified listings across 36 localities, combining Linear Regression, Random Forest, XGBoost, and LightGBM into a single weighted prediction. Computation happens client-side, so no property or personal data is transmitted anywhere. The outcome is a working demonstration of what locally-grounded, explainable property valuation can look like when it's built on actual sales data instead of broker discretion.

Problem Statement:- A buyer or investor evaluating a property in Jammu today has no real way to test whether the asking price holds up. Appraisals are slow and costly, brokers are financially motivated to push prices upward, and no free tool exists that reflects what comparable properties have actually sold for in that specific locality. The result is that meaningful capital moves on incomplete, one-sided information, with no way for the buyer to check the claim independently before committing. That's the gap and it's also the opening: transparent, instant valuation for a market like this doesn't currently exist, which is exactly the space this project occupies.

Objectives and Scope:- This phase covers price prediction across residential and commercial property types using 35 input features; an investment score built from location quality, growth indicators, and legal standing; locality-level exploration across all 36 covered areas, including historical and forecast price movement; and a comparison view for evaluating up to five properties side by side, backed by a model layer that explains its own reasoning rather than functioning as a black box. Deliberately out of scope for now: live listing feeds, financing or loan-eligibility checks, real-time transaction updates, and coverage beyond Jammu. Those belong to the next phase.

Deliverable:-

- A prediction engine that returns a price estimate and investment score specific to the property entered, not a generic citywide average.
- A single exploration view replacing the current mix of broker quotes, scattered listings, and guesswork.
- A complete evaluation flow - prediction, locality context, comparison, explanation that takes a user from raw property details to a defensible decision.
- Documented model performance by property type and price band, so the accuracy claims are checkable rather than assumed.
- A dataset and modelling layer that states its own boundaries clearly, giving later work something solid to extend rather than a black box to reverse-engineer.

Future Scope:- The same modelling approach extends naturally to other tier-II Indian cities facing the same lack of transparent local pricing tools. Loan-eligibility and rental-yield estimation would add real value given how many users are financing rather than buying outright, and incorporating property photos alongside the existing structured features would sharpen predictions for properties that don't fit the standard mould.